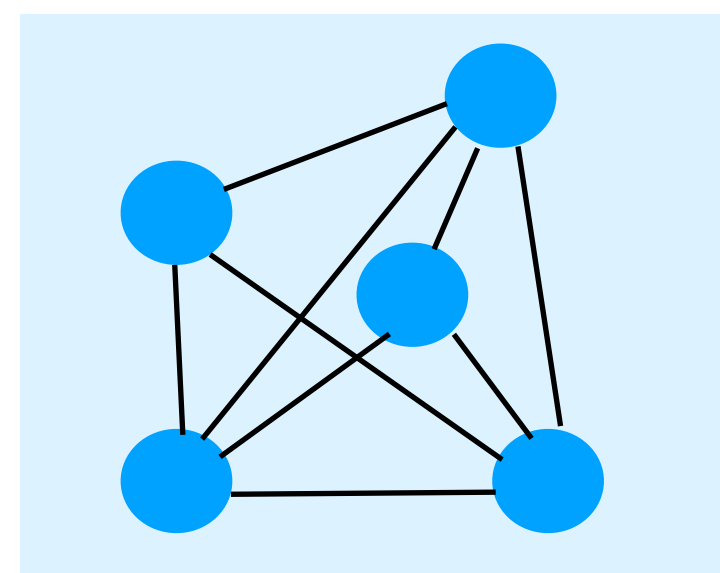


Graph Learning and Node Subset Selection under Sparsity Constraint

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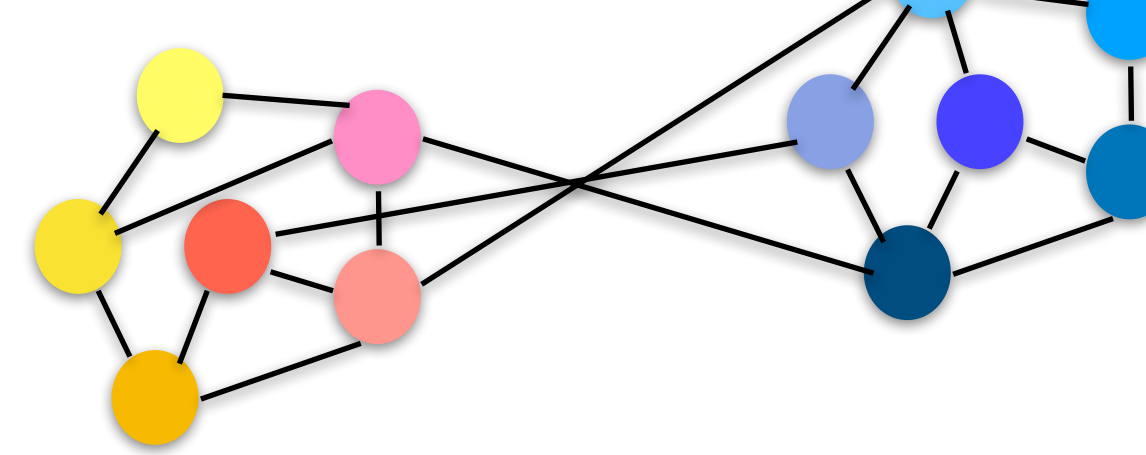
Graph Learning



Based on the **Measurements** recorded across Entities present in a Population Set, the **Idea** is to **infer the Underlying Graphical Relation** among them.

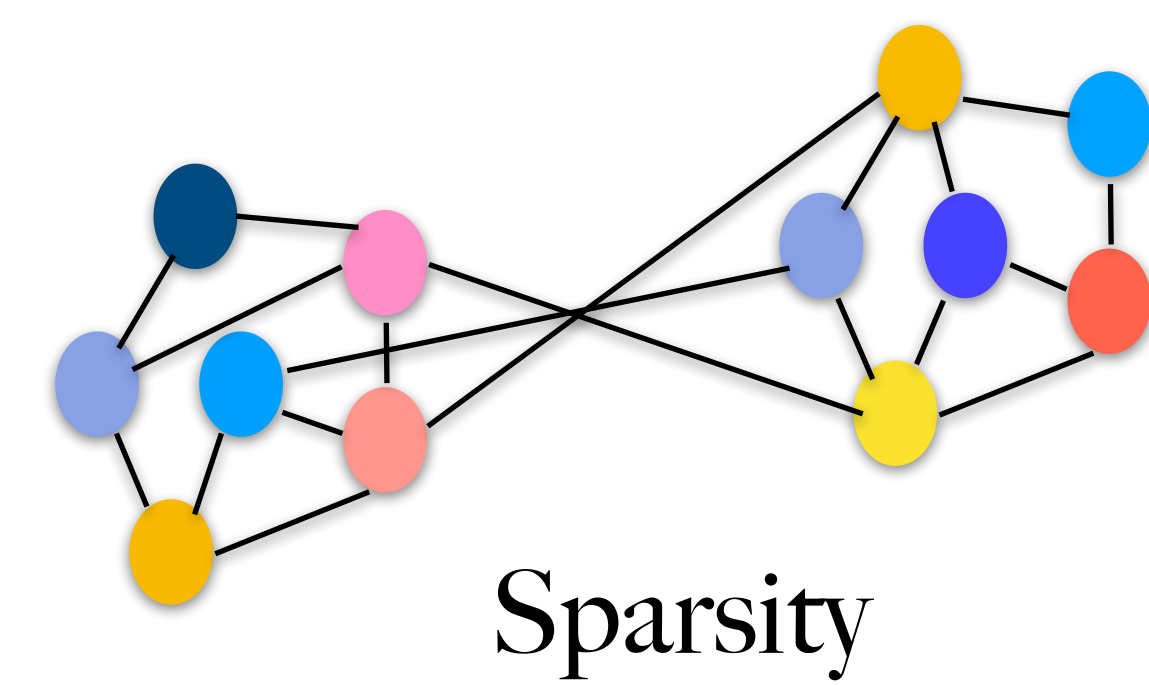
	T1	T2	T3	T4	T5	T6
1	Red	Green	Magenta	Blue	Green	Red
2	Magenta	Yellow	Blue	Red	Orange	Magenta
3	Blue	Cyan	Magenta	Cyan	Blue	Red
4	Green	Red	Yellow	Magenta	Red	Cyan

Smoothness



If Data

Says so



Sparsity

Under Sparsity Constraint

Traditionally '**Smoothness**' i.e. Near-By Entities should possess Similar Attributes, **was an Assumption**. But to be more Pragmatic, an Evolution of Measurements recorded over a population need **not** be resulted from a Smooth **Diffusion Process**.

Node Subset Selection via Compressed Sensing CS

Let $X \in \mathbb{R}^M$ and $Y \subseteq X$
 $\text{size}(Y) = K$

C be a Subset Selector Matrix

$$Y = C \cdot X$$

Based on the subset Y , we want to reconstruct X
Sparse Signal Reconstruction

Optimise for $X \quad \|Y - C \cdot X\|_2^2$

But this has **Infinitely** Many Solutions

So how could we obtain X uniquely?

How can CS help in this?

CS takes the advantage of the fact that many signals are **Sparse** i.e. they contain many coefficients close to or equal to zero, when represented in **some Domain**.

$$X \xrightarrow{\text{Transform}} S$$

Using some Transforming Matrix ψ

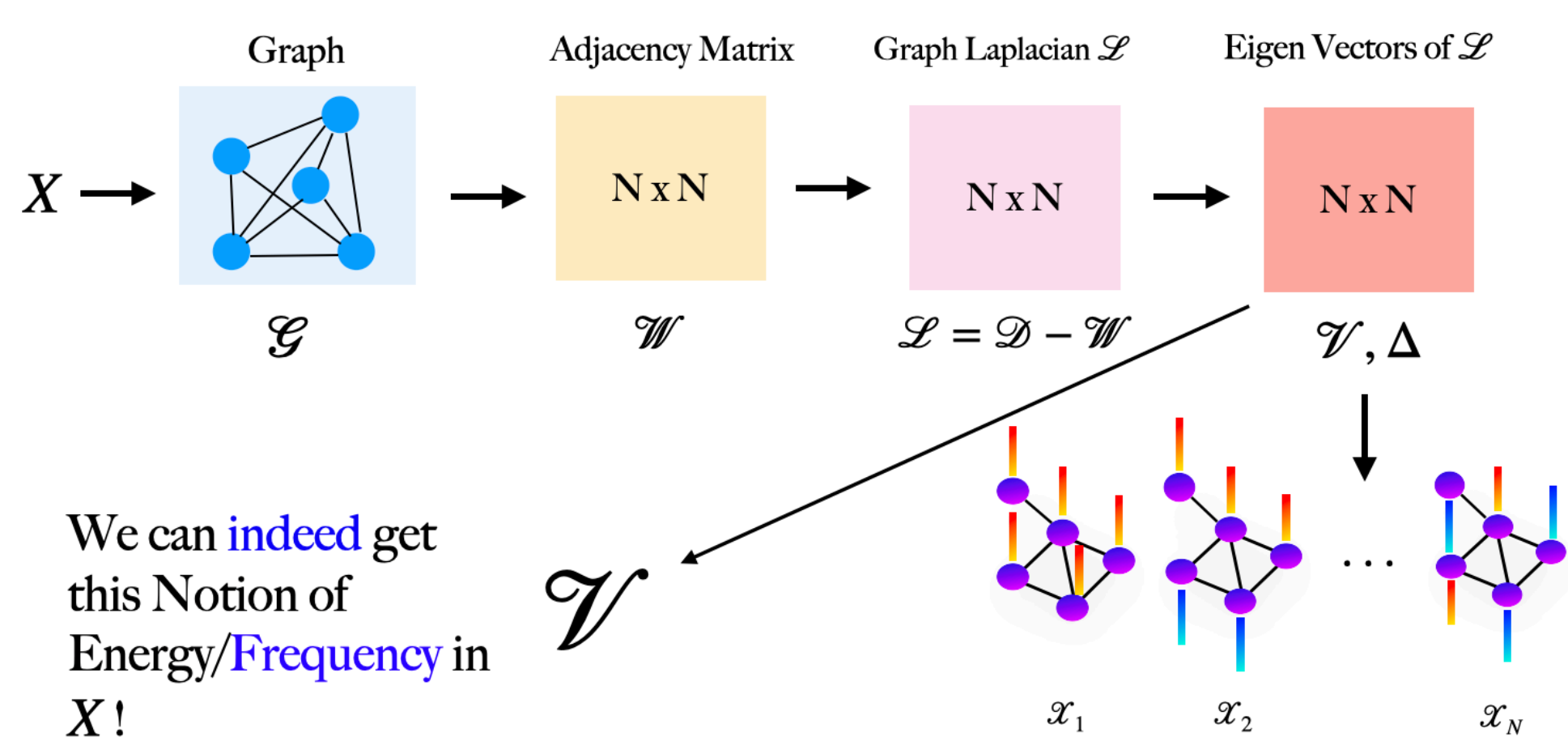
$$X \cdot \psi = s$$

Law of Conservation of Energy $\Rightarrow$ Total Energy in this System should be Preserved!

$$X = \psi^{-1} \cdot s$$

$\Rightarrow$ Components of ψ^{-1} must Capture **Notion of Energy/Frequency**

Now from where we can get this Notion of Energy/Frequency in X ?



We can **indeed** get this Notion of Energy/Frequency in X !

$$\text{So } \psi^{-1} = V \Rightarrow X = V \cdot s$$

Recall Subset Selection Problem that we began with

$$Y = C \cdot X \Rightarrow C \cdot V \cdot s$$

But CS says something more...

CS says, Not all Under-Determined System of Linear Equations have a Sparse Solution. However if there exists a **Unique Sparse Solution** to the Under-Determined System, then the Compressed Sensing Framework allows the recovery of that Solution.

So, $X = V \cdot s$ i.e. Exact Recovery Only possible when s is the sparsest

So for Nice Recovery We propose the Following

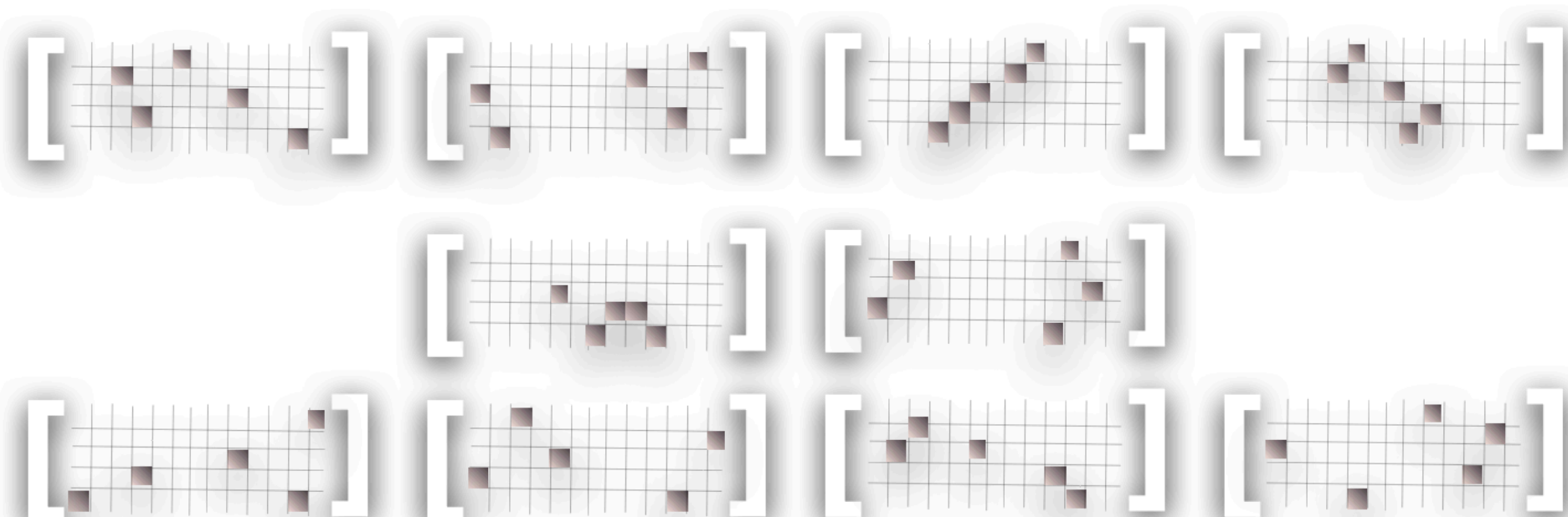
$$s^* = \min_s \|Y - C \cdot V \cdot s\|_2^2 + \lambda \cdot \|s\|_1$$

Recovery

$$X^* = V \cdot s^*$$

Performance of this Model **depends** upon the Choice of Subset, so **Subset Selection Matrix** Needs a Honourable Mention.

Subset Selector Matrix C 101



So there are Variety of Way to Select a Subset

Let's Try To Perform **Subset Selection** to Efficiently Handle

Measles Outbreak in Mumbai, Nov'22

Step 0. What is Measles ?

A DISEASE WHICH SPREADS EASILY FROM PERSON TO PERSON THROUGH AIR, AND CAN CAUSE FEVER, COUGH, RASH.

Step 1. Analyse the Data Set

Example Setting Considers Dataset from Measles dataset from BMC

Name of Patient	Address/Village	Hospital Registered With	Ward	Measles Y/N
Person 1	Colaba	JJ Hospital	A	Y
Person 2	Antop Hill	Cooper Hospital	B	N
Person 3	Gowardi	Kasturba Hospital	M	Y
Person 4	Andheri	Lilavati Hospital	L	Y

Step 2. We Calculated How many cases were Registered From Village or at a Hospital.

Hospital	# Cases	Lat/Long	Village	# Cases	Lat/Long
H1	123	19'E, 24N	V1	20	19'E, 24N
H2	45	19'E, 24N	V2	125	19'E, 24N
H3	13	19'E, 24N	V3	25	19'E, 24N

Every Entity in Population Related with a Scalar Signal Value i.e. #Cases
Let there be N Entities in Total. $X \in \mathbb{R}^N$, is a Vector of Health Stats of Entities
Entities $\in \{H, V\}$

Step 3. Graph Based Interpretation of Setting Developed in Last Step

Now we all know very well why Is $\mathcal{V}$ important?

Adjacency Matrix $\mathcal{W}$ (N x N) $\rightarrow$ Graph Laplacian $\mathcal{L}$ (N x N) $\rightarrow$ Eigen Vectors of $\mathcal{L}$ (N x N) $\mathcal{V}, \Delta$

Step 4. Compressed Sensing Model for Optimal Reconstruction

For obtaining a **Reconstruction** of the Health Status over the **Entire Population** we need to **Solve** the Following, and obtain the s

$$\min_s \|Y - C \cdot \mathcal{V} \cdot s\|_2^2 + \lambda \cdot \|s\|_1$$

Select an Appropriate Subset Selector Matrix C

$$\min_s \|Y - C \cdot \mathcal{V} \cdot s\|_2^2 + \lambda \cdot \|s\|_1$$

Now we can solve the above optimisation problem, to obtain s^*
*Hyperparameter λ is obtained via Cross-Validation

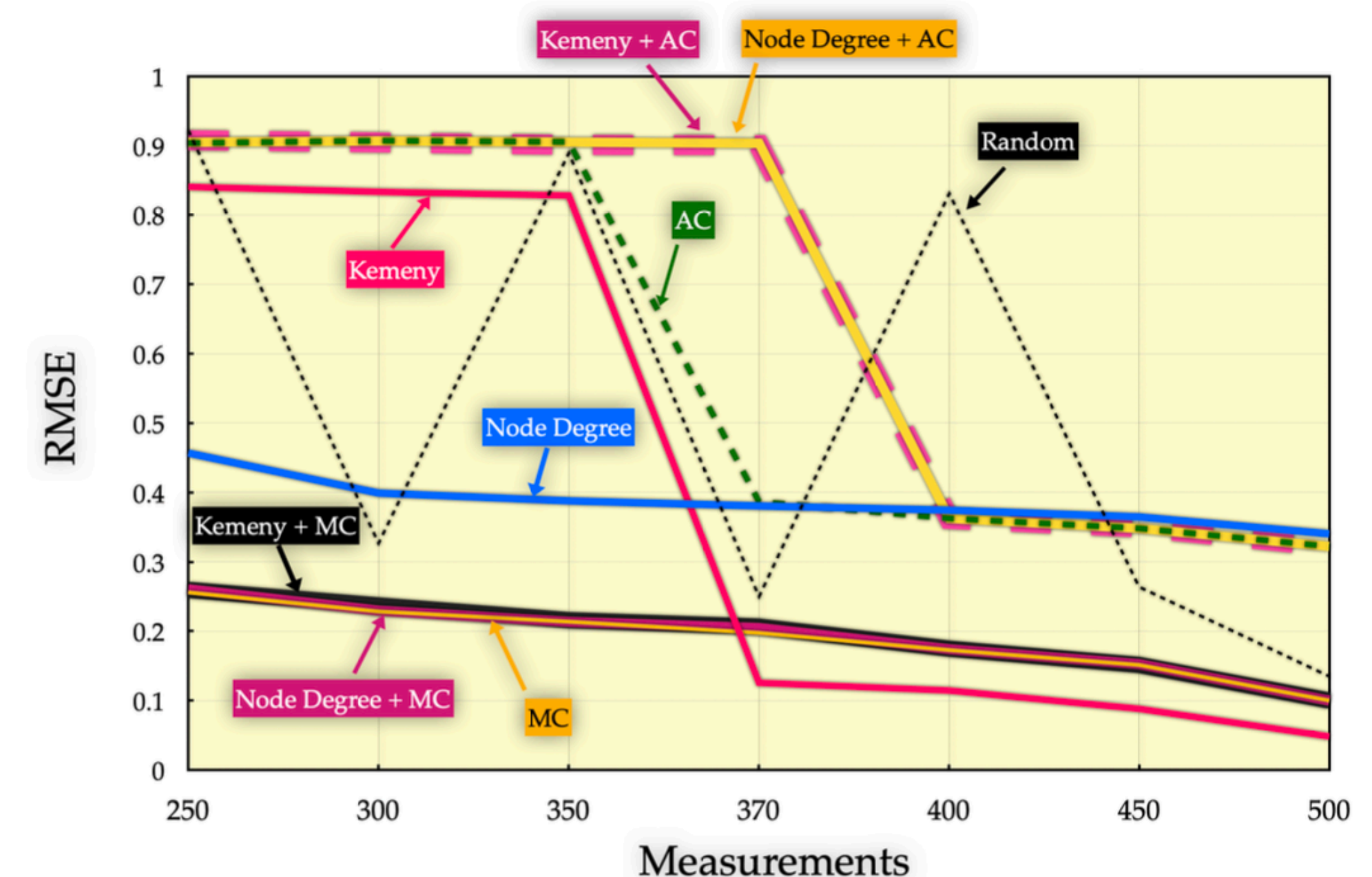
$$X^* = \mathcal{V} \cdot s^* \quad \text{Reconstruction Obtained}$$

Objective of WHY we are DOING, WHATEVER we are DOING

When comprehensive data across regions is lacking, can we still make accurate predictions using available data? Our approach is to **identify a representative subset** from the available regions and use their data to forecast for the entire set.

To tackle this challenge, we've developed a model inspired by **Sparse Signal Reconstruction Problem**, akin to **Compressed Sensing**. This model helps us **determine the optimal subset** of regions with available data, facilitating accurate data interpolation across other regions **leveraging inter-regional connectivity data**.

The Figure on the top right helps us judge the efficacy of the model we built. This plot shows the quality of forecasts obtained when interpolated with various subset selection strategies governed by the subset selector matrix C . **It also shows that about 75-80% of the total regions got a correct prediction when interpolated with the best subset selection strategy.**



Next, We aim to **enhance our predictions** by learning **richer** and representative **graphs** from the available data for each region. This effort will **empower** us to execute **downstream tasks** such as **subset selection** and **interpolation**, even more **precisely**.

So we started analysing the **literature** to see how the graphs (more precisely the relation i.e. edge-weights among the entities in the fixed population set) are being learnt and thus we found that **when the edge weights are not naturally defined by an application, one common way to define the weight of an edge connecting vertices i and j , $W_{i,j}$ is via a thresholded Gaussian kernel weighting function, $W_{i,j} = e^{-\gamma d_{i,j}^2}$, where $d_{i,j}$ represents distance between 2 Entities.**

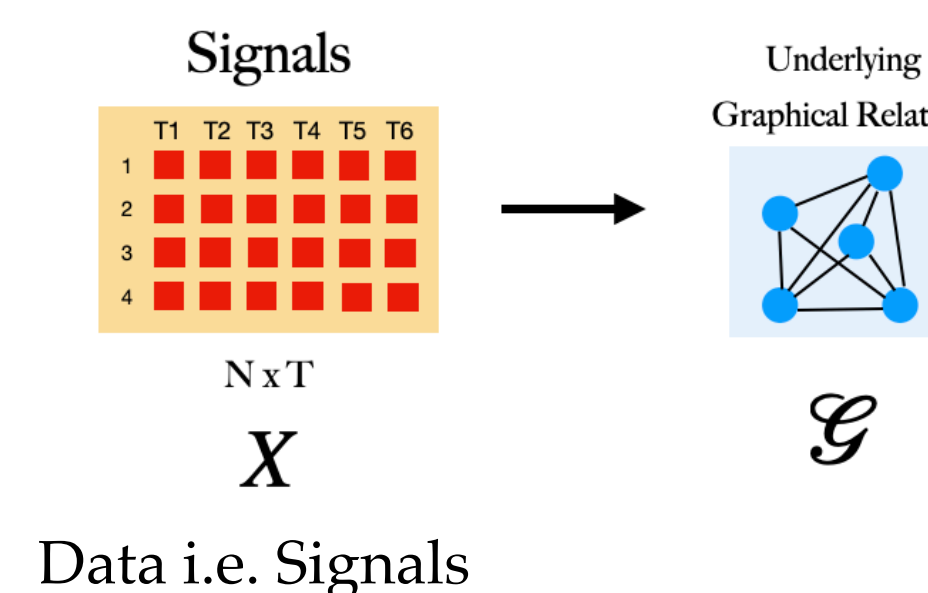
This was a very ubiquitous heuristic applied to infer the edge weight of a Graph. But we wanted totally data driven way to 'LEARN' the Graphs through the **worth of available Data**. So we started digging into **fully Data-Driven** ways to infer the the graph, we noticed that in these techniques, a prior of specific kind (Band-limitedness i.e. **Smoothness** i.e. Data which changes smoothly over near-by Entities in a Population) on Data was being considered. **Ideally this could have been True, if all the signals were Generated from a pure Diffusion Process, but which is certainly not Pragmatic!**

Thus we propose Our method which learns the Graph, using the Data available in **data-driven way** and which **doesn't consider such priors** on the Data.

Problem Formulation

Given T observations for N Entities, we would like to infer the Underlying **Relationship** that Exists within them.

Each Column of the X Matrix Represents a Graph Signal defined over all the Entities.



Proposed Formulation to Learn the Graph given the Data

$$\min_{\alpha, \Delta, V} \|X - V \cdot \alpha\|_F^2 + \mu_1 \cdot \|\alpha\|_{1,1} + \mu_2 \cdot \|V \Delta V^T\|_{1,1} + \mu_3 \cdot \|\hat{\Theta} - V \Delta V^T - \frac{1}{\sigma^2} \cdot I\|_F^2$$

$$\text{s.t.} \quad \begin{cases} V^T V = I_N, & (a) \\ \frac{1}{\sqrt{N}} \mathbf{1}_N^T \cdot V = [100 \dots 0], & (b) \\ \Delta = \text{diag}(0, \lambda_2, \dots, \lambda_N) \geq 0, & (c) \\ V \Delta V^T \cdot \mathbf{1}_N = \mathbf{0}_N, & (d) \\ \text{diag}(V \Delta V^T) \geq 0, & (e) \\ V \Delta V^T - \text{diag}(\text{diag}(V \Delta V^T)) \leq 0, & (f) \end{cases}$$

Notes on the Proposed Algorithm:

- We do **not** assume Band-limitedness **prior** on the **Data** while learning the underlying Graph.
- Precisely speaking we are learning the **Eigen Vectors and the Eigen Values of the Graph Laplacian L** , so that $V \cdot \Delta \cdot V^T = L$. From L , we infer the **Graph** i.e. the Adjacency Matrix W as $W = D - L$, where D is a diagonal Matrix whose diagonal elements contains the Degrees of the Entities in the Population. Interesting point, the Degrees can be obtained from the learnt Laplacian itself, as they sit on its diagonals. $D_{i,i} = L_{i,i}$.

Future Research Prospects that should naturally follow this work includes:

- Sample Complexity Analysis:** What amount of Measurements are Necessarily required for Learning a Good Graph?
- Compressed Sensing applied on to Graphs.**
- What will happen if **some more entities** gets **added** in the Population, how will then the **topology** then change?

Algorithm 1 Our Graph Learning Algorithm

Input: $X \in \mathbb{R}^{N \times T}$, σ_{oracle} $\triangleright$ Signals i.e. X : T Observations across N Entities in a Population
Output: α^* , Δ^* , V^*
Initialization:
[V, Δ, U] $\leftarrow$ SVD(X)
 $v_1 \leftarrow \frac{1}{\sqrt{N}} \mathbf{1}_N$ $\triangleright$ Make first column of V as $\frac{1}{\sqrt{N}} \mathbf{1}_N$
 $V \leftarrow \text{QR}(V)$
 $\mu \leftarrow \text{mean}(X)$ $\triangleright \mu \in \mathbb{R}^{N \times 1}$: Mean of observations over each Node
 $\hat{\Theta} \leftarrow \left(\frac{(X - \mu)(X - \mu)^T}{T-1} \right)^{-1}$ $\triangleright \hat{\Theta} \in \mathbb{R}^{N \times N}$: Precision Matrix or Inverse Covariance Matrix
 $\sigma = \sigma_{\text{oracle}}$
while not convergence do
 α -step: Learning latent representation of the Signals in the Fourier Space.
 $\min_{\alpha} \|X - V \cdot \alpha\|_F^2 + \mu_1 \cdot \|\alpha\|_{1,1}$
 Δ -step: Learning Eigen Values of the Graph Laplacian L .
 $\min_{\Delta} \mu_2 \cdot \|V \Delta V^T\|_{1,1} + \mu_3 \cdot \|\hat{\Theta} - V \Delta V^T - \frac{1}{\sigma^2} \cdot I\|_F^2$
 s.t. $\begin{cases} \Delta = \text{diag}(0, \lambda_2, \dots, \lambda_N) \geq 0, & (a) \\ V \Delta V^T \cdot \mathbf{1}_N = \mathbf{0}_N, & (b) \\ \text{diag}(V \Delta V^T) \geq 0, & (c) \\ V \Delta V^T - \text{diag}(\text{diag}(V \Delta V^T)) \leq 0, & (d) \end{cases}$
 V -step: Learning Eigen Vectors of the Graph Laplacian L .
 $\min_V \|X - V \cdot \alpha\|_F^2 + \mu_2 \cdot \|V \Delta V^T\|_{1,1} + \mu_3 \cdot \|\hat{\Theta} - V \Delta V^T - \frac{1}{\sigma^2} \cdot I\|_F^2$
 s.t. $\begin{cases} V^T V = I_N, & (a) \\ \frac{1}{\sqrt{N}} \mathbf{1}_N^T \cdot V = [100 \dots 0], & (b) \\ V \Delta V^T \cdot \mathbf{1}_N = \mathbf{0}_N, & (c) \\ \text{diag}(V \Delta V^T) \geq 0, & (d) \\ V \Delta V^T - \text{diag}(\text{diag}(V \Delta V^T)) \leq 0, & (e) \end{cases}$
end while